

Class 8: Breast Cancer Mini-Project

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##Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a data-set from fine needle aspiration (FNA) of breast mass.

Data Import

The data is made available for you to download. We can read this in using `read.csv()`:

```
wisc.df <- read.csv("WisconsinCancer.csv")
```

```
# Save your input data file into your Project directory
```

```
fna.data <- "WisconsinCancer.csv"
```

```
# Complete the following code to input the data and store as wisc.df
```

```
wisc.df <- read.csv(fna.data, row.names=1)
```

```
head(wisc.df, 3)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean	
842302	M	17.99	10.38	122.8	1001	
842517	M	20.57	17.77	132.9	1326	
84300903	M	19.69	21.25	130.0	1203	
		smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302		0.11840	0.27760	0.3001	0.14710	
842517		0.08474	0.07864	0.0869	0.07017	
84300903		0.10960	0.15990	0.1974	0.12790	
		symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302		0.2419	0.07871	1.0950	0.9053	8.589
842517		0.1812	0.05667	0.5435	0.7339	3.398
84300903		0.2069	0.05999	0.7456	0.7869	4.585
		area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302		153.40	0.006399	0.04904	0.05373	0.01587
842517		74.08	0.005225	0.01308	0.01860	0.01340
84300903		94.03	0.006150	0.04006	0.03832	0.02058
		symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302		0.03003	0.006193	25.38	17.33	
842517		0.01389	0.003532	24.99	23.41	
84300903		0.02250	0.004571	23.57	25.53	
		perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302		184.6	2019	0.1622	0.6656	
842517		158.8	1956	0.1238	0.1866	
84300903		152.5	1709	0.1444	0.4245	
		concavity_worst	concave.points_worst	symmetry_worst		
842302		0.7119	0.2654	0.4601		
842517		0.2416	0.1860	0.2750		
84300903		0.4504	0.2430	0.3613		
		fractal_dimension_worst				
842302		0.11890				
842517		0.08902				
84300903		0.08758				

Make sure we remove or exclude the **diagnosis** column from the data-set that we use for further analysis - this is the expert diagnosis as either M or B:

```
# We can use -1 here to remove the first column
wisc.data <- wisc.df[,-1]
```

```
# Create diagnosis vector for later
diagnosis <- as.factor( wisc.df$diagnosis )
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(diagnosis)
```

```
diagnosis  
  B    M  
357 212
```

OR

```
sum (wisc.df$diagnosis == "M" )
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```
length(grep("_mean", colnames(wisc.data) ))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
# Perform PCA on wisc.data by completing the following code
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335

	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966

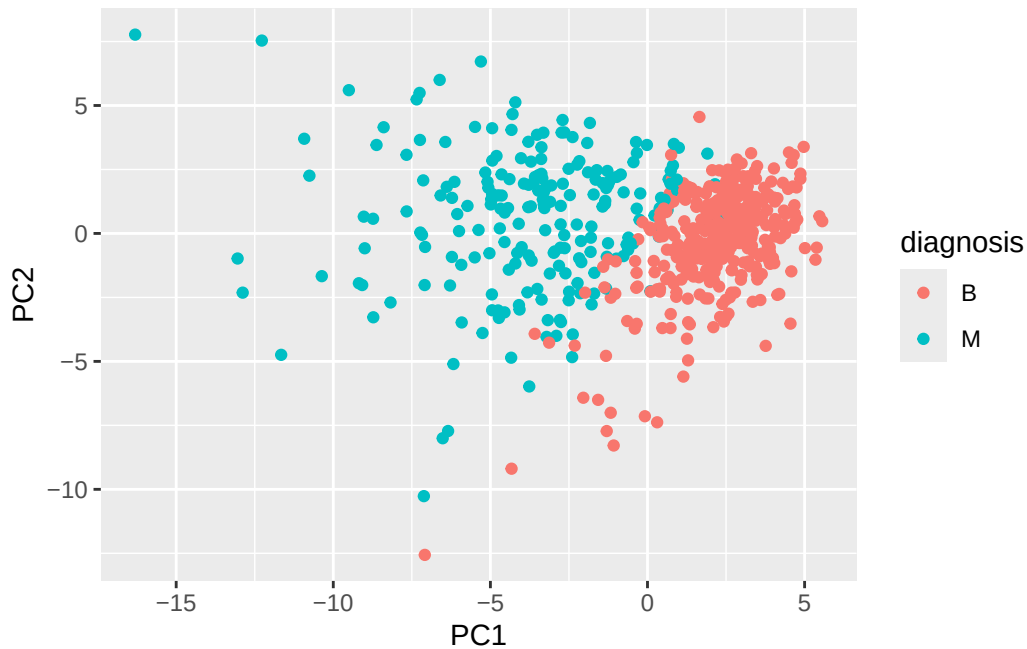
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997

	PC29	PC30
Standard deviation	0.02736	0.01153
Proportion of Variance	0.00002	0.00000
Cumulative Proportion	1.00000	1.00000

Let's see the "PC score plot"

```
library(ggplot2)

ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 = 0.4427; PC1 captures 44.27% of the variance

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

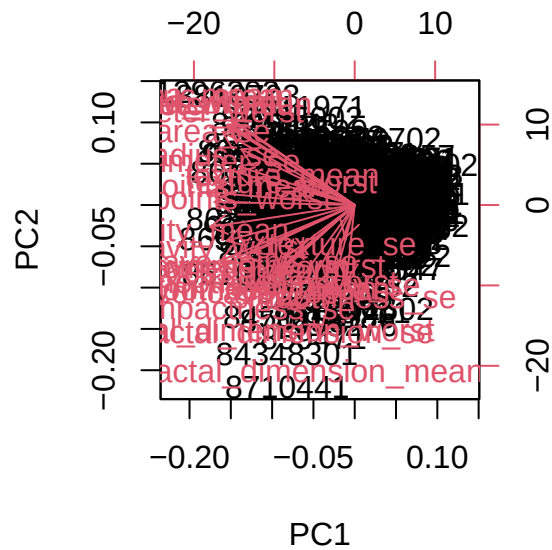
Look at cumulative proportion: PC1 = 0.4427, PC2 = 0.6324, PC3 = 0.72636. 3 principal components

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

Look at cumulative: PC6 = 0.8876 (not enough), PC7 = 0.9101; 7 principal components

Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

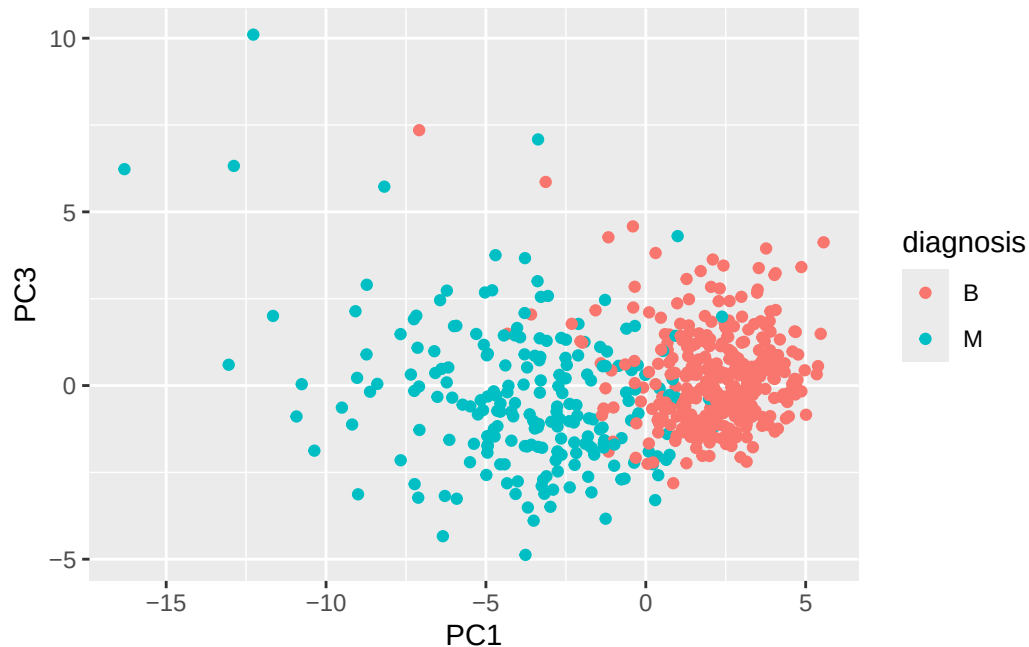
```
biplot(wisc.pr)
```



The plot is very cluttered and difficult to interpret because there are too many points and variable labels overlapping. It is hard to clearly see patterns or relationships between variables and observations. It's a hot mess; not an effective visualization for this data-set.

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
ggplot(wisc.pr$x) +
  aes(PC1, PC3, col=diagnosis) +
  geom_point()
```



The separation between benign (B) and malignant (M) samples is mainly along PC1, while PC3 does not add much additional separation. The two groups still overlap more compared to the PC1 vs PC2 plot, indicating that PC2 captures more meaningful variation for distinguishing the classes than PC3.

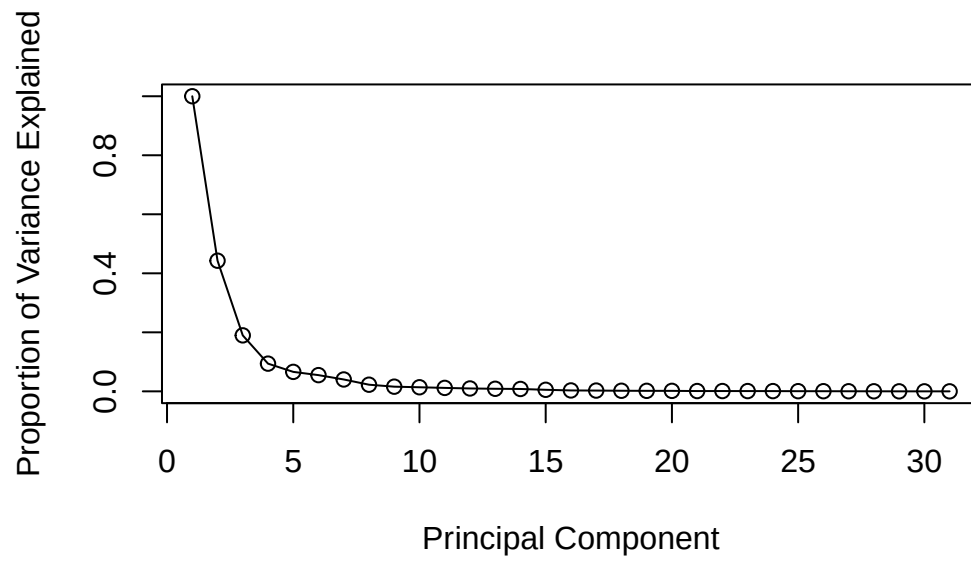
#Variance Explained

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

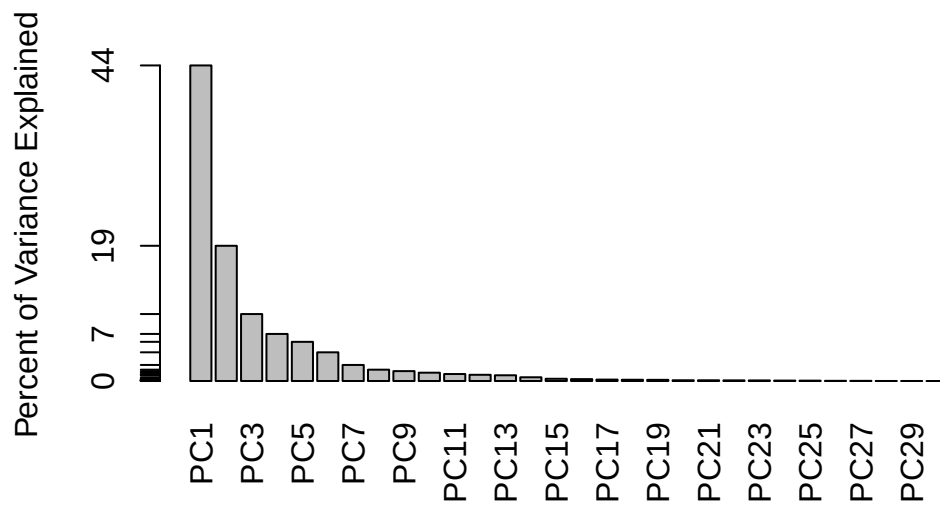
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- (wisc.pr$sdev^2) / sum(wisc.pr$sdev^2)

# Plot variance explained for each principal component
plot(c(1,pve), xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```

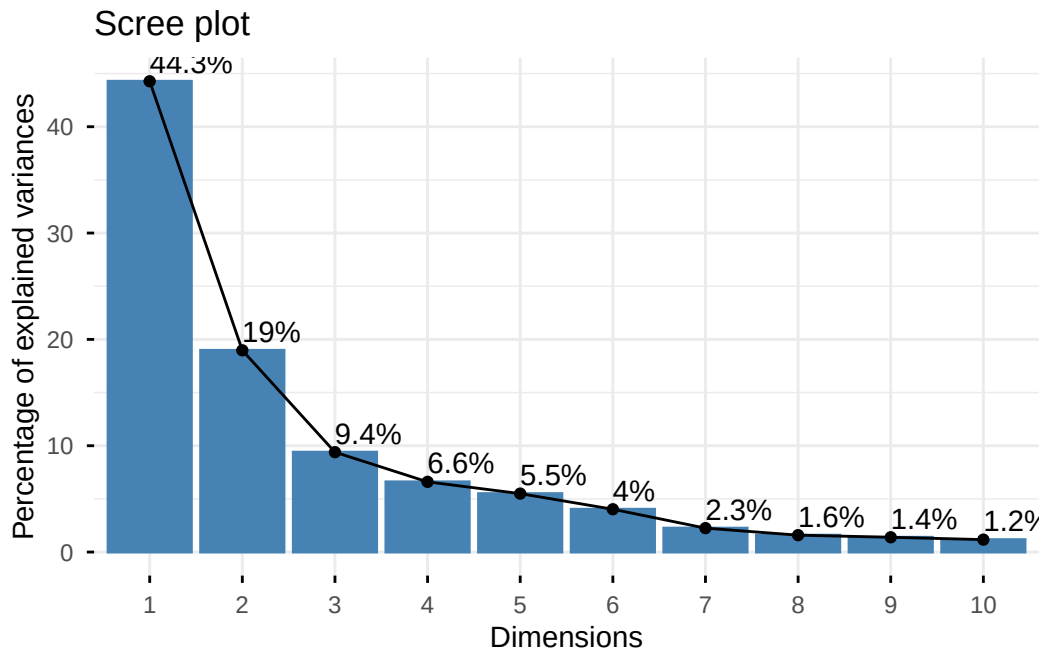



```
## ggplot based graph
#install.packages("factoextra")
library(factoextra)
```

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/en/product-guide-to-principal-component-methods-in-r/>

```
fviz_eig(wisc.pr, addlabels = TRUE)
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean", 1]
```

```
[1] -0.2608538
```

The loading value for `concave.points_mean` on PC1 is large and positive (around ~ 0.26), indicating it strongly contributes to PC1. Yes, there are features with slightly larger contributions, such as `radius_mean`, `perimeter_mean`, and `area_mean`, which have even higher loading values.

Hierarchical clustering

```
str(wisc.data)
```

```
'data.frame': 569 obs. of 30 variables:
 $ radius_mean      : num  18 20.6 19.7 11.4 20.3 ...
 $ texture_mean     : num  10.4 17.8 21.2 20.4 14.3 ...
 $ perimeter_mean   : num  122.8 132.9 130 77.6 135.1 ...
 $ area_mean        : num  1001 1326 1203 386 1297 ...
 $ smoothness_mean  : num  0.1184 0.0847 0.1096 0.1425 0.1003 ...
 $ compactness_mean : num  0.2776 0.0786 0.1599 0.2839 0.1328 ...
 $ concavity_mean   : num  0.3001 0.0869 0.1974 0.2414 0.198 ...
 $ concave.points_mean : num  0.1471 0.0702 0.1279 0.1052 0.1043 ...
 $ symmetry_mean    : num  0.242 0.181 0.207 0.26 0.181 ...
 $ fractal_dimension_mean : num  0.0787 0.0567 0.06 0.0974 0.0588 ...
 $ radius_se        : num  1.095 0.543 0.746 0.496 0.757 ...
 $ texture_se       : num  0.905 0.734 0.787 1.156 0.781 ...
 $ perimeter_se     : num  8.59 3.4 4.58 3.44 5.44 ...
 $ area_se          : num  153.4 74.1 94 27.2 94.4 ...
 $ smoothness_se    : num  0.0064 0.00522 0.00615 0.00911 0.01149 ...
 $ compactness_se   : num  0.049 0.0131 0.0401 0.0746 0.0246 ...
 $ concavity_se     : num  0.0537 0.0186 0.0383 0.0566 0.0569 ...
 $ concave.points_se : num  0.0159 0.0134 0.0206 0.0187 0.0188 ...
 $ symmetry_se      : num  0.03 0.0139 0.0225 0.0596 0.0176 ...
 $ fractal_dimension_se : num  0.00619 0.00353 0.00457 0.00921 0.00511 ...
 $ radius_worst     : num  25.4 25 23.6 14.9 22.5 ...
 $ texture_worst    : num  17.3 23.4 25.5 26.5 16.7 ...
 $ perimeter_worst  : num  184.6 158.8 152.5 98.9 152.2 ...
 $ area_worst       : num  2019 1956 1709 568 1575 ...
 $ smoothness_worst : num  0.162 0.124 0.144 0.21 0.137 ...
 $ compactness_worst : num  0.666 0.187 0.424 0.866 0.205 ...
 $ concavity_worst  : num  0.712 0.242 0.45 0.687 0.4 ...
 $ concave.points_worst : num  0.265 0.186 0.243 0.258 0.163 ...
 $ symmetry_worst   : num  0.46 0.275 0.361 0.664 0.236 ...
 $ fractal_dimension_worst: num  0.1189 0.089 0.0876 0.173 0.0768 ...
```

```
sum(is.na(wisc.data))
```

```
[1] 0
```

```
wisc.data <- na.omit(wisc.data)
```

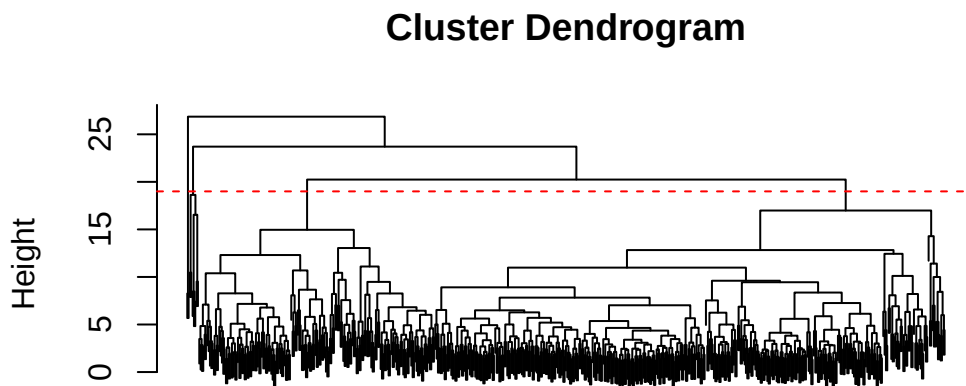
```
# Scale the wisc.data data using the "scale()" function
data.scaled <- scale(wisc.data)
```

```
data.dist <- dist(data.scaled)
```

```
wisc.hclust <- hclust(data.dist, method = "complete")
```

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust, labels = FALSE)  
abline(h = 19, col="red", lty=2)
```



```
data.dist  
hclust (*, "complete")
```

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
```

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Q11. OPTIONAL: Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 6? How do you judge the quality of your result in each case?

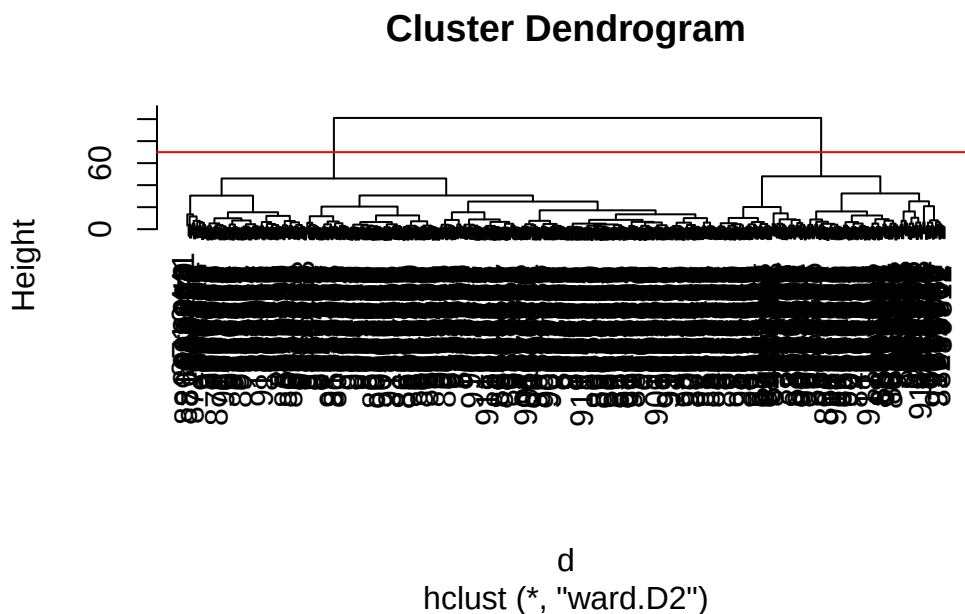
2 clusters works best since it matches the two diagnoses (B vs M). More clusters reduce clarity.

12. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

The ward.D2 method gives the best results because it produces more clearly separated and compact clusters that better match the benign (B) and malignant (M) diagnoses. This method minimizes variance within clusters, leading to more meaningful and well-defined groupings compared to single, complete, or average linkage.

Combining Methods

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```



```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1  2
171 398
```

How does this clustering `grps` correspond to the expert diagnosis?

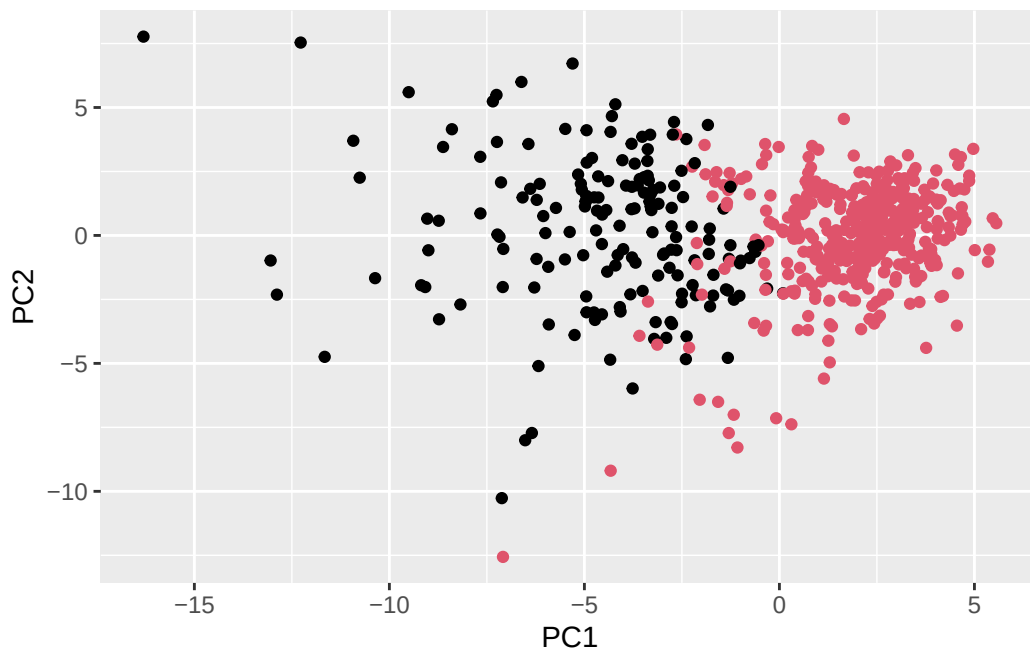
```
table(diagnosis)
```

```
diagnosis
  B  M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1  2
  B      6 351
  M     165  47
```

```
ggplot(wisc.pr$x) +
  aes(PC1, PC2) +
  geom_point(col=grps)
```



```
## Use the distance along the first 7 PCs for clustering i.e. wisc.pr$x[, 1:7]
wisc.pr.hclust <- hclust(dist(wisc.pr$x[,1:7]), method="ward.D2")
```

```
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)

table(wisc.pr.hclust.clusters, diagnosis)
```

```

      diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24
```

Q13. How well does the newly created hclust model with two clusters separate out the two “M” and “B” diagnoses?

```
# Compare to actual diagnoses
table(wisc.pr.hclust.clusters, diagnosis)
```

```

      diagnosis
wisc.pr.hclust.clusters  B  M
1      28 188
2     329  24
```

The clustering performs well. One cluster is mostly malignant (188 M vs 28 B), and the other is mostly benign (329 B vs 24 M), showing good separation with only a small number of misclassifications.

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

```
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

The clustering without PCA performs worse. It produces more mixed clusters, with less clear separation between malignant and benign samples compared to the PCA-based clustering. PCA improves clustering by reducing noise and focusing on the most important variation.

Sensitivity/ Specificity

Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

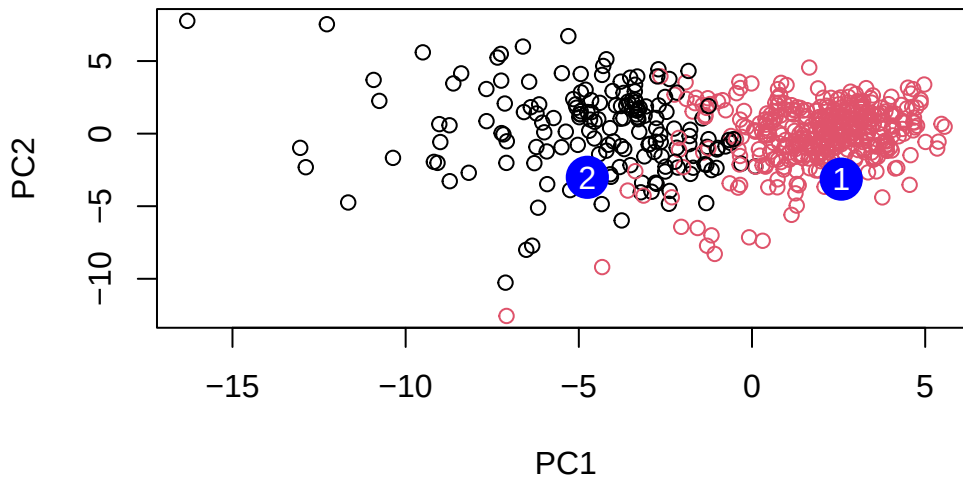
The PCA-based clustering model resulted in the best specificity and sensitivity. It more accurately separates benign and malignant samples, leading to fewer false positives and false negatives compared to clustering on the original data.

Prediction

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```


	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2 lies closer to the malignant cluster in the PCA plot, indicating a higher likelihood of being cancerous, and should therefore be prioritized for follow-up. Patient 1 appears closer to the benign group.